Modification

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Double linked list

1) Problem statement:
To develop a student Record Management system using a doubly linked list in C++. The system should allow users to insert, display, search, delete and update student records, sort records according to marks in descending order, and display records in reverse order. The system maintains both previous and next pointers to support traversal in both directions and efficient record management.

2) Objectives:
- To implement a doubly linked list in C++.
- To store student roll number, name and marks.
- To insert and display student records.
- To search for a student using roll number.
- To delete and update student records.
- To sort records by marks in descending order.
- To display records in reverse order.
- To understand how prev and next pointers work.

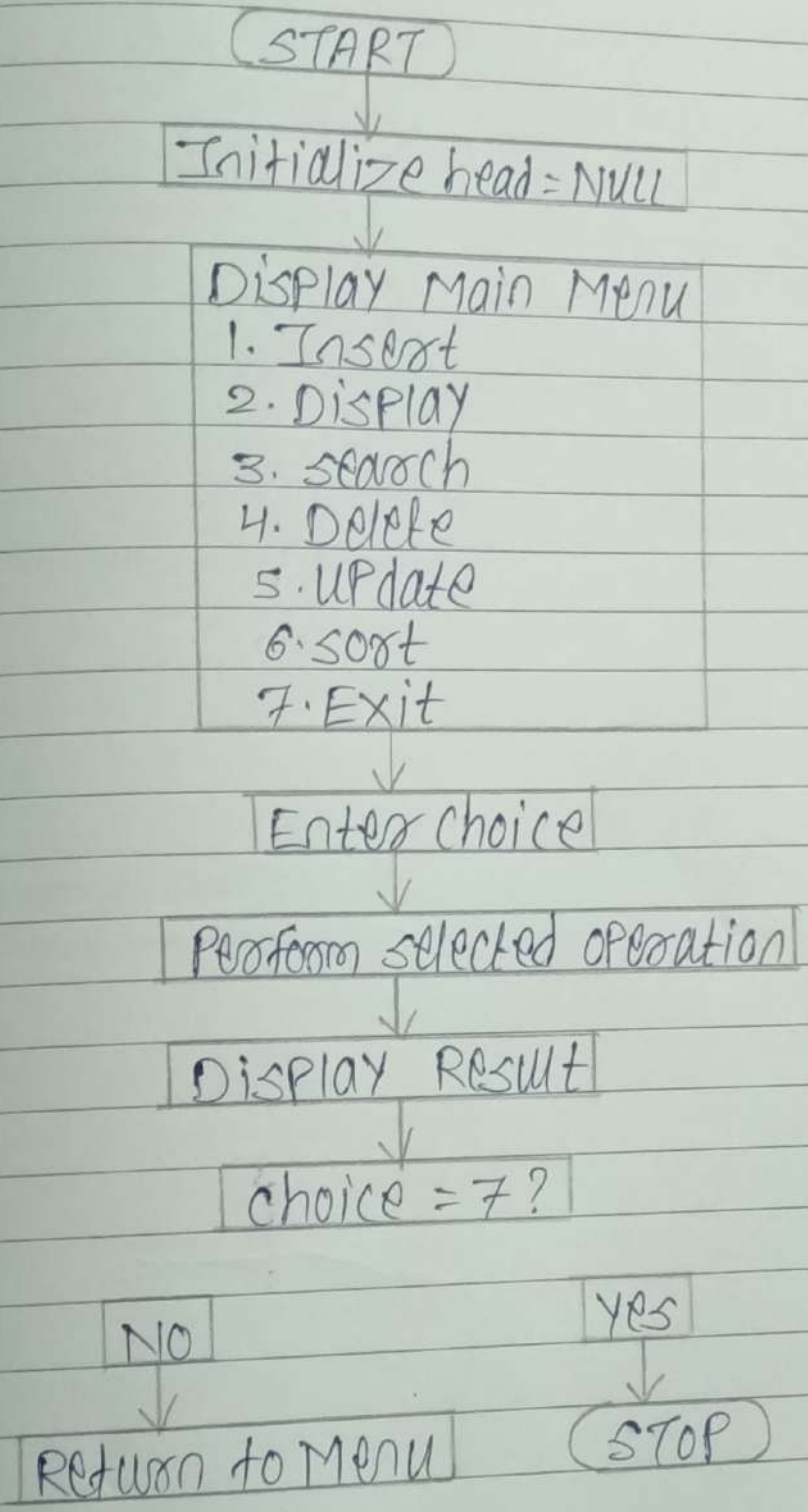
3) Algorithm:
1) Start
2) Initialize head = Null and tail = Null.
3) Display the menu of operations.
4) Accept the user's choice.
5) Perform the selected operation:

- Insert: Create a new node and connect it to the last node using Prev and next.
- Display: Traverse from head to tail and disp all records.
- search: Find the record using the student's roll number.
- Delete: Find the student and adjust both Pre and next links before deleting the node.
- Update: change the name and marks of the selected student.
- sort: Arrange student records by marks from highest to lowest.
- Reverse Display: Traverse from tail to head
- Exit: Terminate the Program.

6) Display the result.
7) If the choice is not 8, return to menu
8) stop.

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Flowchart:



```
#include <iostream>
#include <string>
using namespace std;

struct Node {
   int roll;
   string name;
   float marks;
   Node *prev, *next;
};

Node *head = NULL, *tail = NULL;

// 1. Insert record
void insert() {
   Node *n = new Node;

   cout << "Enter roll, name and marks: ";
   cin >> n->roll >> n->name >> n->marks;

   n->prev = tail;
   n->next = NULL;

   if (head == NULL)
      head = tail = n;
   else {
      tail->next = n;
      tail = n;
   }

   cout << "Record inserted\n";
}
```

```
        // 2. Display records forward
        void display() {
          Node *t = head;

          if (t == NULL) {
            cout << "No records found\n";
            return;
          }

        while (t != NULL) {
          cout << t->roll << " "
              << t->name << " "
              << t->marks << endl;
        t = t->next;
      }
    }

    // 3. Search record
    void search() {
      int r;
      cout << "Enter roll number: ";
      cin >> r;

    Node *t = head;

    while (t != NULL) {
      if (t->roll == r) {
        cout << "Found: " << t->name
            << ", Marks: " << t->marks << endl;
        return;
    }
    t = t->next;
```

```
    }
    cout << "Record not found\n";
}

// 4. Delete record
void deleteRecord() {
    int r;
    cout << "Enter roll number to delete: ";
    cin >> r;

    Node *t = head;

    while (t != NULL && t->roll != r)
        t = t->next;

    if (t == NULL) {
        cout << "Record not found\n";
        return;
    }

    if (t->prev != NULL)
        t->prev->next = t->next;
    else
        head = t->next;

    if (t->next != NULL)
        t->next->prev = t->prev;
    else
        tail = t->prev;

    delete t;
    cout << "Record deleted\n";
```

```
    }

    // 5. Update record
    void update() {
      int r;
      cout << "Enter roll number to update: ";
      cin >> r;

      Node *t = head;

      while (t != NULL && t->roll != r)
        t = t->next;

      if (t == NULL) {
        cout << "Record not found\n";
        return;
      }

      cout << "Enter new name and marks: ";
      cin >> t->name >> t->marks;

      cout << "Record updated\n";
}

// 6. Sort by marks (descending)
void sortMarks() {
  for (Node *i = head; i != NULL; i = i->next) {
    for (Node *j = i->next; j != NULL; j = j->next) {
      if (i->marks < j->marks) {
        swap(i->roll, j->roll);
        swap(i->name, j->name);
        swap(i->marks, j->marks);
      }
```

```
    }
  }
  cout << "Records sorted by marks\n";
}

// 7. Display records in reverse
void reverseDisplay() {
  Node *t = tail;

  if (t == NULL) {
    cout << "No records found\n";
    return;
  }

  while (t != NULL) {
    cout << t->roll << " "
       << t->name << " "
       << t->marks << endl;
    t = t->prev;
  }
}

int main() {
  int ch;

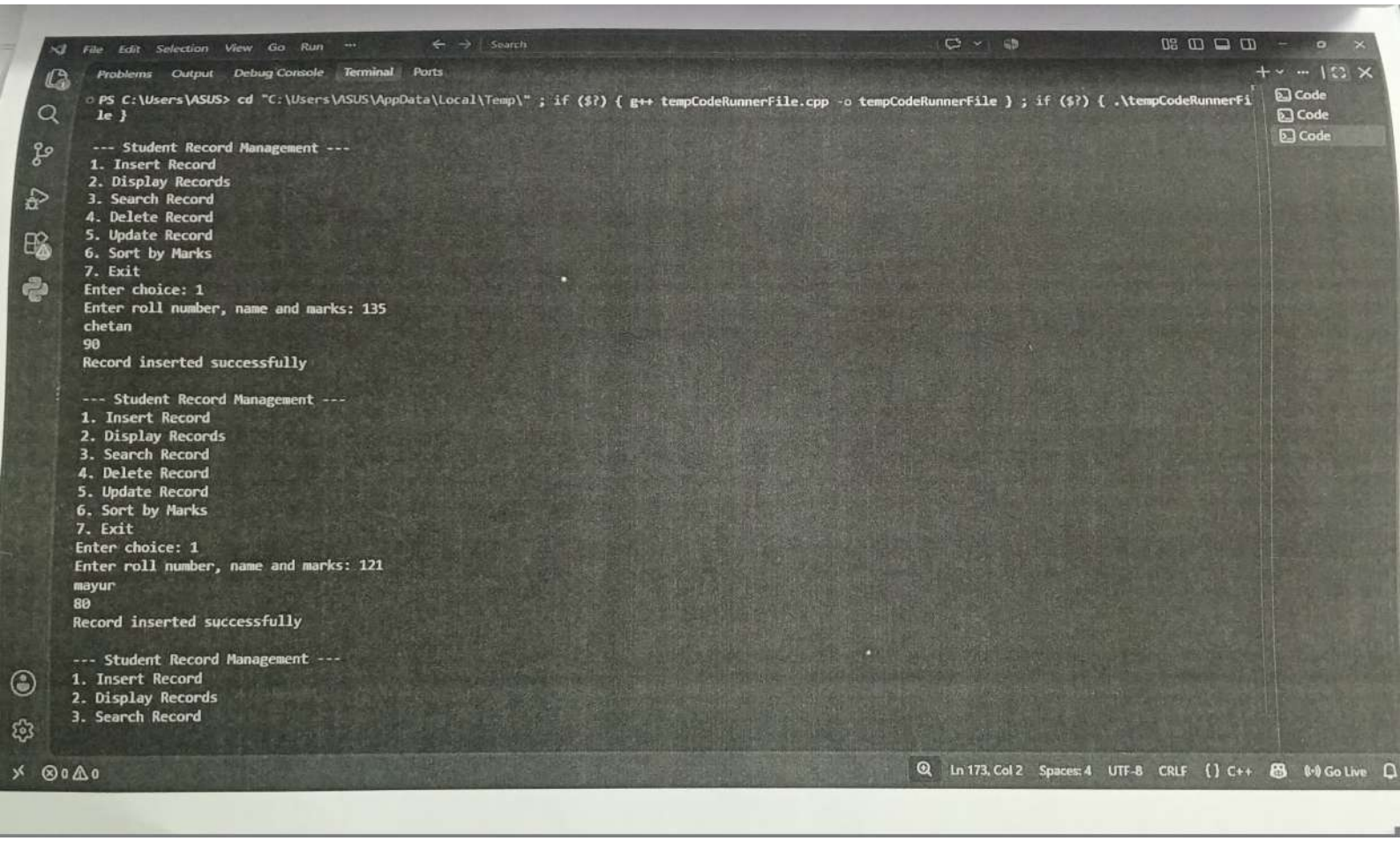
  do {
    cout << "\n--- Student Record System ---\n";
    cout << "1. Insert Record\n";
    cout << "2. Display Records\n";
    cout << "3. Search Record\n";
    cout << "4. Delete Record\n";
    cout << "5. Update Record\n";
```

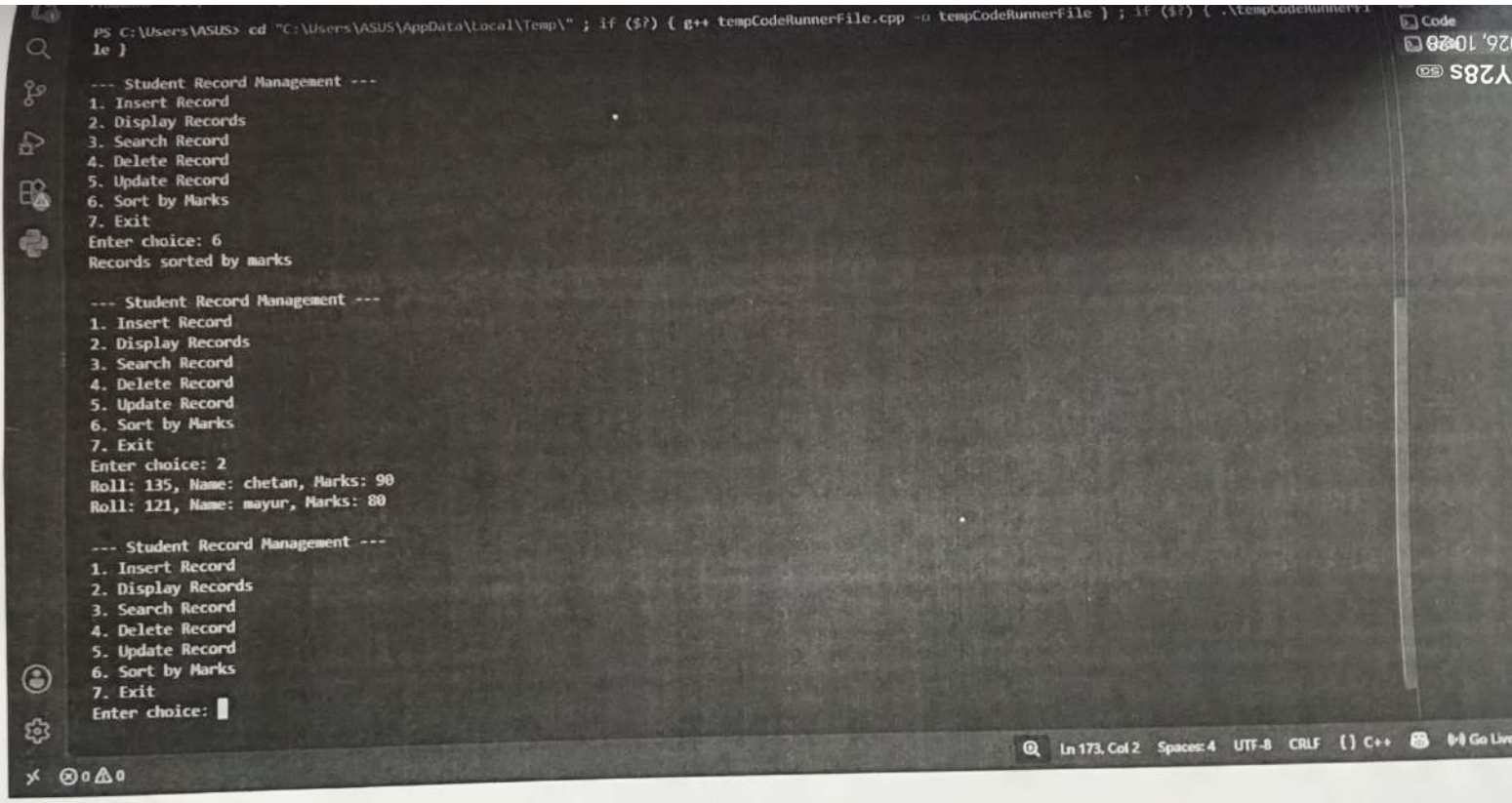
```
        cout << "6. Sort by Marks\n";
        cout << "7. Reverse Display\n";
        cout << "8. Exit\n";
        cout << "Enter choice: ";
        cin >> ch;

        switch (ch) {
            case 1: insert(); break;
            case 2: display(); break;
            case 3: search(); break;
            case 4: deleteRecord(); break;
            case 5: update(); break;
            case 6: sortMarks(); break;
            case 7: reverseDisplay(); break;
            case 8: cout << "Program terminated\n"; break;
            default: cout << "Invalid choice\n";
        }
    } while (ch != 8);

    // Free remaining nodes
    while (head != NULL) {
        Node *t = head;
        head = head->next;
        delete t;
    }
    tail = NULL;

    return 0;
}
```


File Edit Selection View Go Run
Search
Problems Output Debug Console Terminal Ports
PS C:\Users\ASUS> cd "C:\Users\ASUS\AppData\Local\Temp\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
--- Student Record Management ---
1. Insert Record
2. Display Records
3. Search Record
4. Delete Record
5. Update Record
6. Sort by Marks
7. Exit
Enter choice: 1
Enter roll number, name and marks: 135
chetan
90
Record inserted successfully
--- Student Record Management ---
1. Insert Record
2. Display Records
3. Search Record
4. Delete Record
5. Update Record
6. Sort by Marks
7. Exit
Enter choice: 1
Enter roll number, name and marks: 121
mayur
80
Record inserted successfully
--- Student Record Management ---
1. Insert Record
2. Display Records
3. Search Record
Code
Code
Code
Ln 173, Col 2
Spaces: 4
UTF-8
CRLF
C++
Go Live


PS C:\Users\ASUS> cd "C:\Users\ASUS\AppData\Local\Temp\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
--- Student Record Management ---
1. Insert Record
2. Display Records
3. Search Record
4. Delete Record
5. Update Record
6. Sort by Marks
7. Exit
Enter choice: 6
Records sorted by marks
--- Student Record Management ---
1. Insert Record
2. Display Records
3. Search Record
4. Delete Record
5. Update Record
6. Sort by Marks
7. Exit
Enter choice: 2
Roll: 135, Name: chetan, Marks: 90
Roll: 121, Name: mayur, Marks: 80
--- Student Record Management ---
1. Insert Record
2. Display Records
3. Search Record
4. Delete Record
5. Update Record
6. Sort by Marks
7. Exit
Enter choice:
Ln 173, Col 2
Spaces: 4
UTF-8
CRLF
C++
Go Live
0 0